EEE-101
Self study: Feb 22, 2026

$$G(s) = \frac{B(s)}{A(s)} \quad \begin{matrix} \rightarrow \text{degree } m \\ \rightarrow \text{degree } n \end{matrix}$$

$\downarrow$

Transfer function

$n \geq m$
- $\rightarrow$ strictly proper $(n > m)$
- $\rightarrow$ Bi-proper $(n = m)$

} Proper

$n < m \rightarrow$ improper

degree of a transfer function is equal to the no. of poles of the transfer function.

* Note that if a pole and zero lie on the same point, they cancel out each other.

$$u(t) \longrightarrow \boxed{G(s) = \frac{s+\alpha}{s+\beta}} \longrightarrow y(t)$$

$$Y(s) = U(s)G(s) = \frac{(s+\alpha)}{s(s+\beta)} = \frac{A}{s} + \frac{B}{s+\beta}$$

$$A = \alpha/\beta, \quad B = \frac{\alpha-\beta}{-\beta} = 1 - \frac{\alpha}{\beta}$$

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$$Y(s) = \frac{(\alpha/\beta)}{s} + \frac{(1-\alpha/\beta)}{s+\beta}$$

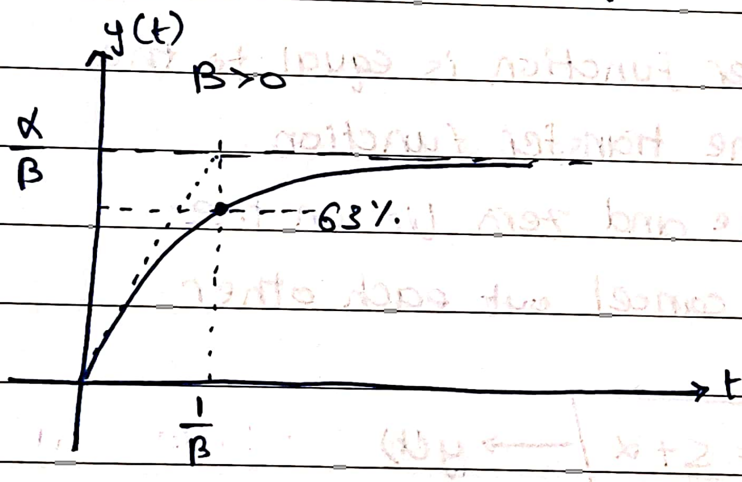
$$y(t) = \underbrace{\frac{\alpha}{\beta}u(t)}_{\text{Forced Response}} + \underbrace{\left(1-\frac{\alpha}{\beta}\right)e^{-\beta t}u(t)}_{\text{Natural Response}}$$

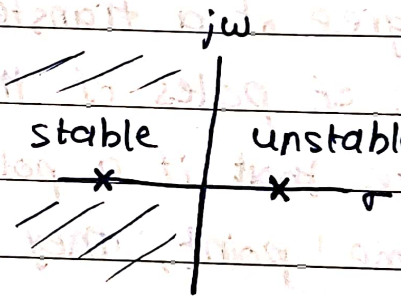
→ pole location

$\beta > 0 \Rightarrow$ stable system

$\beta < 0 \Rightarrow$ unstable system

$\beta = 0 \Rightarrow$ system is an integrator (marginaly stable)





↳ time constant $= \frac{1}{|\text{pole location}|} = \tau$

Rise time (t_r) : time to go from 10% to 90% of the steady state value.

Settling time (t_s) : time for response to reach and stay within 2% of maximum value.

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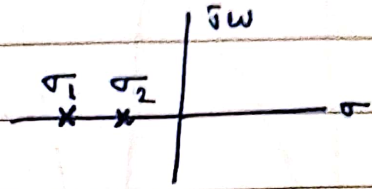
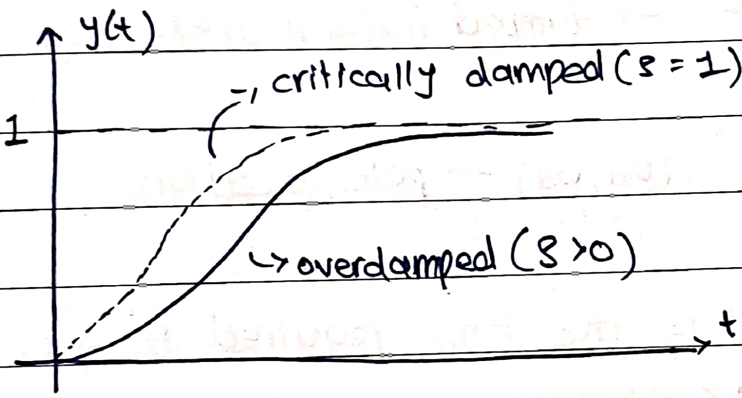
$$u(t) \longrightarrow \boxed{G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}} \longrightarrow y(t)$$

Poles : $-\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$



$\zeta > 1 \Rightarrow$ Real distinct poles (overdamped)

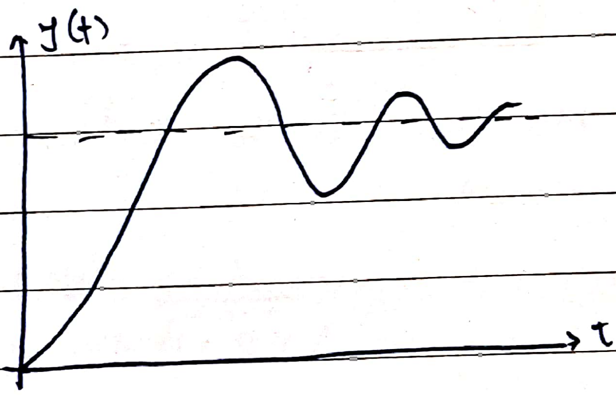
$\zeta = 1 \Rightarrow$ Real repeated poles (critically damped)

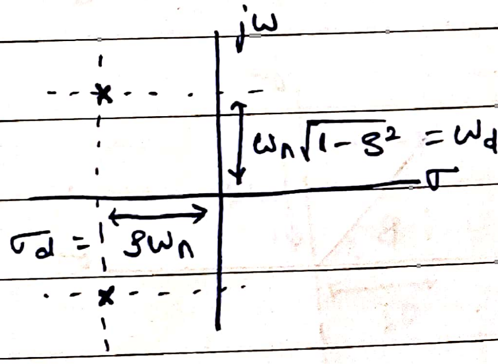


ζ : damping ratio

ω_n : natural frequency

$0 < \zeta < 1 \Rightarrow$ Complex conjugate poles (underdamped)





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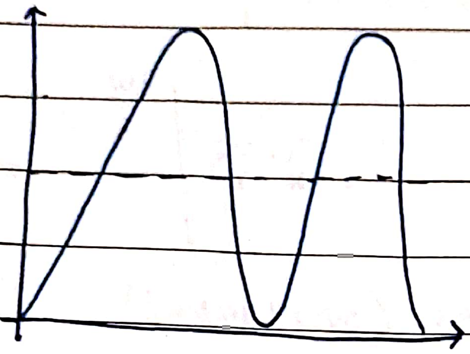
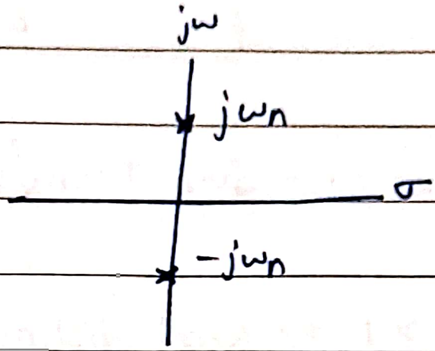
$\xi = 0 \Rightarrow$ Inversely imaginary roots (undamped)





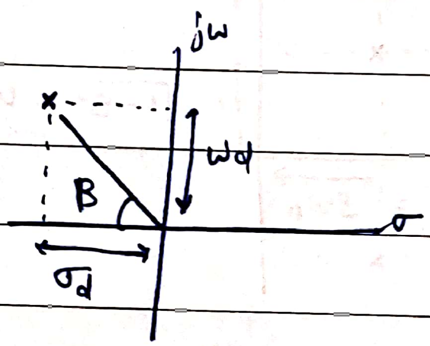
$\omega_d = \omega_n\sqrt{1-\xi^2}$ → damped natural frequency

$\xi < 1$

$\sigma_d = \xi\omega_n$ (σ_d, ω_d) → pole location

Rise time (t_r): It is the time required to go from 10% to 90% in case of overdamped and time required to go from 0% to 100% for underdamped.

$$t_r = \frac{1}{\omega_d}\tan^{-1}\left(\frac{-\omega_d}{\sigma_d}\right) = \frac{1}{\omega_d}(\pi - \beta)$$



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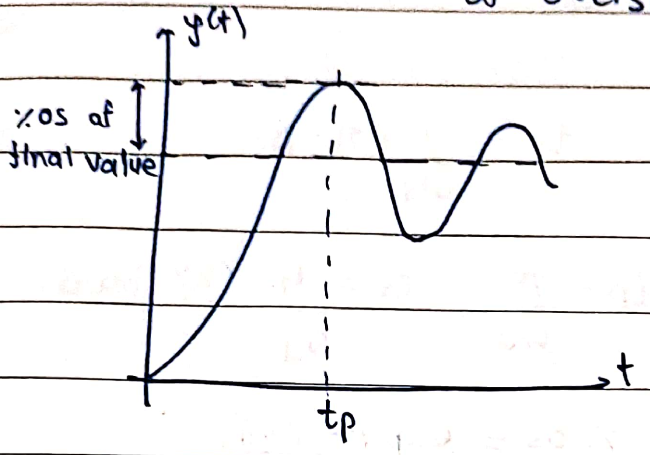
Peak time (t_p) : Time to reach the first peak of overshoot.



$$\left.\frac{dy(t)}{dt}\right|_{t=t_p} = 0$$

$$t_p = \frac{\pi}{\omega_d}$$

$$\%\ \text{overshoot} = \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right) = \exp\left(\frac{-\pi\sigma_d}{\omega_d}\right)$$

$$\text{settling time } (t_s) = \frac{-\ln\left(0.02\sqrt{1-\zeta^2}\right)}{\zeta\omega_n} \cong \frac{4}{\zeta\omega_n} \cong \frac{4}{\sigma_d}$$

$$y(t) = 1 - \frac{e^{-\overset{\sigma_d}{\cancel{\zeta\omega_n}}\, t}}{\sqrt{1-\zeta^2}} \cos(\omega_d t - \phi)$$

$$= 1 - \frac{e^{-\overset{\sigma_d}{\cancel{\zeta\omega_n}}\, t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \psi)$$

$$\phi = \tan^{-1}\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right), \quad \psi = \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$

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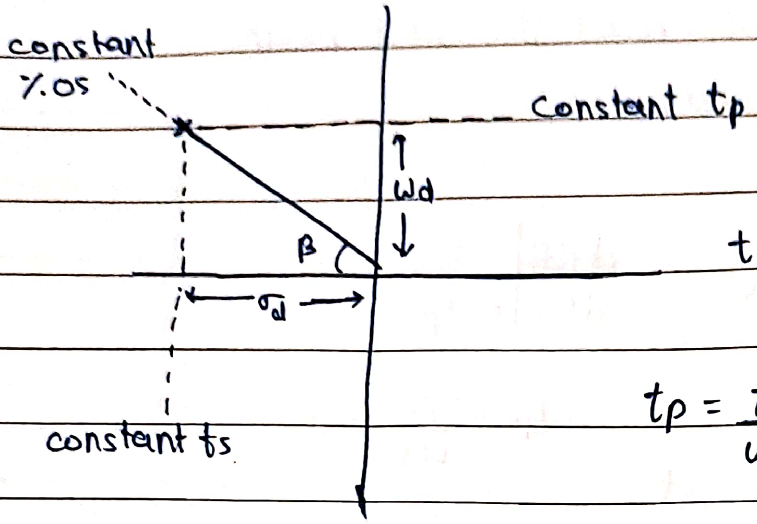
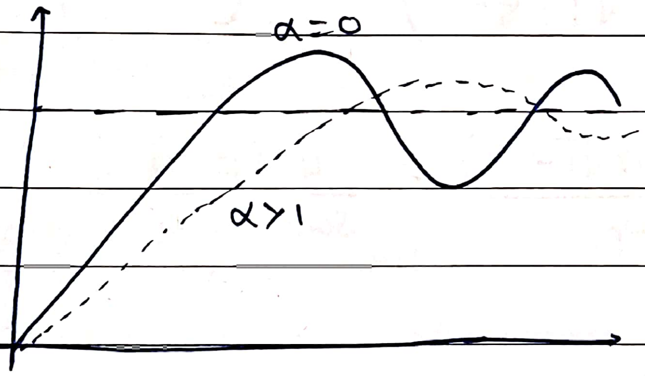


$$t_r = \frac{1}{\omega_d}(\pi - \beta)$$

$$t_p = \frac{\pi}{\omega_d}, \quad t_s = \frac{4}{\sigma_d} \ (2\% \text{ band})$$

$$\% \, OS = \exp\left(\frac{-\pi \sigma_d}{\omega_d}\right)$$

Effect of adding poles: $G(s) = \dfrac{\omega_n^2}{(\alpha s+1)(s^2 + 2\zeta\omega_n s + \omega_n^2)}$

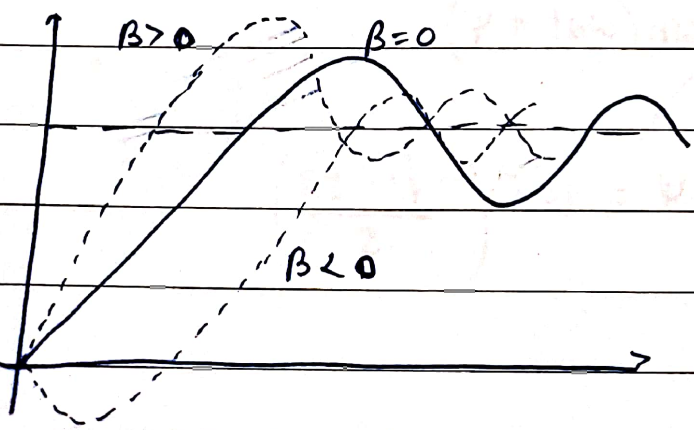


→ smoothens curve

→ increases rise time

→ decreases overshoot

Effect of adding zeroes: $G(s) = \dfrac{(\beta s+1)\,\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$



→ decreases rise time

→ increases overshoot

→ causes undershoot if $\beta < 0$

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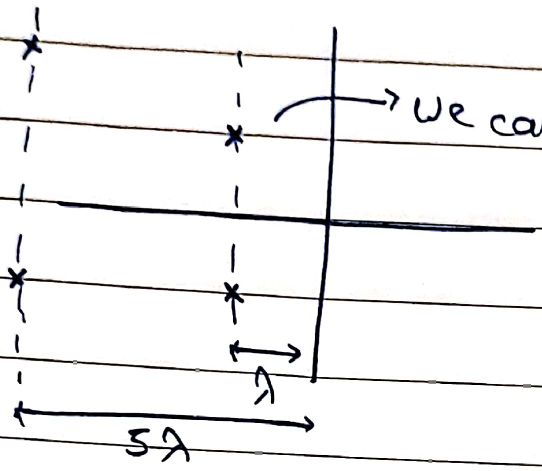
A group of poles are dominant if they are at a distance of more than 5 times from other poles distance.



→ We can ignore the effect of these poles.